

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
C _{ADC}	Internal sample and hold capacitor	-	-	3	-	pF
t _{ADCVREG_STUP}	ADC LDO startup time	-	-	-	10	μs
t _{STAB}	ADC power-up time	LDO already started	1	-	-	conversion cycle
t _{OFF_CAL}	Offset calibration time	-	85			1/f _{ADC}
t _{LATR}	Trigger conversion latency for regular and injected channels	Trigger from an asynchronous clock	3	-	4	
		Trigger from a synchronous clock	3			
t _S	Sampling time	-	2.5	-	288.5	
t _{CONV}	Total conversion time (including sampling time)	N-bits resolution	t _S + 1.5 + N			
I _{DDA(ADC)}	ADC consumption on V _{DD} , V _{DDA} and V _{REF}	f _S = 2.25 MSPS	-	200	-	μA

1. High temperature generate leakage current on ADC inputs. This current create voltage drop through R_{AIN} directly affecting ADC accuracy (the worst case when $V_{AIN} = V_{DDA}$). To limit this effect to a tolerance of 2LSBs, it must respect R_{AIN} max tables.

Table 54. 12-bit ADC accuracy

ADC accuracy values are measured after internal calibration. Resolution = 12 bits, no oversampling.

Evaluated by characterization on LQFP64. Packages without a V_{REF} - pad can have degraded specifications. Not tested in production.

Symbol	Parameter	Min	Typ	Max	Unit
ET	Total unadjusted error	-	4	5.7	LSB
EO	Offset error	-	± 2.5	± 5	
EG	Gain error	-	2.6	6	
ED	Differential linearity error	-1	-	1.3	
EL	Integral linearity error	-3	± 2.5	3	
ENOB	Effective number of bits	-	10.7	-	bits
SINAD	Signal-to-noise and distortion ratio	-	66	-	
SNR	Signal-to-noise ratio	-	68	-	dB
THD	Total harmonic distortion	-	-70	-	